DUET ADMISSION: 101 C++ PROGRAMS

১. ক্যালকুলেটর - যোগ, বিয়োগ, গুণ, ভাগ (1. CALCULATOR - SUM SUB MUL DIV (USER INPUT))

```
#include<iostream>
using namespace std;
int main() {
    float a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Sum            = " << a + b << endl;
    cout << "Subtraction    = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    if (b != 0)
        cout << "Division       = " << a / b << endl;
    else
        cout << "Division       = Infinity" << endl;
    return 0;
}
```

২a. ৩টি সংখ্যার মধ্যে বৃহত্তম (2a. LARGEST AMONG 3 NUMBERS)

```
#include<iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a > b && a > c)
        cout << a << " is the largest." << endl;
    else if (b > c)
        cout << b << " is the largest." << endl;
    else
        cout << c << " is the largest." << endl;
    return 0;
}
```

২b. ৩টি সংখ্যার মধ্যে ক্ষুদ্রতম (2b. SMALLEST AMONG 3 NUMBERS)

```
#include<iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a < b && a < c)
        cout << a << " is the smallest." << endl;
    else if (b < c)
        cout << b << " is the smallest." << endl;
    else
        cout << c << " is the smallest." << endl;
    return 0;
}
```

২c. ৩টি সংখ্যার মধ্যে বৃহত্তম ও ক্ষুদ্রতম (উভয়ই) (2c. LARGEST AND SMALLEST AMONG 3 NUMBERS (BOTH))

```
#include<iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a > b && a > c)
        cout << a << " is the largest." << endl;
    else if (b > c)
        cout << b << " is the largest." << endl;
    else
        cout << c << " is the largest." << endl;
    if (a < b && a < c)
        cout << a << " is the smallest." << endl;
    else if (b < c)
        cout << b << " is the smallest." << endl;
    else
        cout << c << " is the smallest." << endl;
    return 0;
}
```

৩. ৩টি সংখ্যার গড় (3. AVERAGE OF 3 NUMBERS)

```
#include<iostream>
using namespace std;
int main() {
    float a, b, c, avg;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    avg = (a + b + c) / 3;
    cout << "Average = " << avg << endl;
    return 0;
}
```

৪a. সমকোণী ত্রিভুজের ক্ষেত্রফল (4a. RIGHT ANGLE TRIANGLE AREA (base * height / 2))

```
#include<iostream>
using namespace std;
int main() {
    float base, height, area;
    cout << "Enter base and height: ";
    cin >> base >> height;
    area = (base * height) / 2;
    cout << "Right Angle Triangle Area = " << area << endl;
    return 0;
}
```

৪b. সমবাহু ত্রিভুজের ক্ষেত্রফল (4b. EQUILATERAL TRIANGLE AREA (sqrt(3)/4 * a^2))

```
    (sqrt(3) = 1.7320508 hard-coded, no math lib)
```

```cpp
#include<iostream>
using namespace std;
int main() {
    float a, area;
    cout << "Enter side length: ";
    cin >> a;
    area = (1.7320508f / 4) * a * a;
    cout << "Equilateral Triangle Area = " << area << endl;
    return 0;
}
```

৪c. বিষমবাহু ত্রিভুজের ক্ষেত্রফল (হেরনের সূত্র) (4c. SCALENE TRIANGLE AREA - HERON'S FORMULA)

```
    (sqrt done manually with Newton's method, no math lib)
```

```cpp
#include<iostream>
using namespace std;
int main() {
    float a, b, c, s, x, area;
    cout << "Enter sides a, b, c: ";
    cin >> a >> b >> c;
    s = (a + b + c) / 2;
    x = s * (s - a) * (s - b) * (s - c);
    // Newton's method sqrt
    float guess = x / 2;
    for (int i = 0; i < 20; i++)
        guess = (guess + x / guess) / 2;
    area = guess;
    cout << "Scalene Triangle Area = " << area << endl;
    return 0;
}
```

৫. বৃত্তের ক্ষেত্রফল ও পরিধি (5. CIRCLE AREA & CIRCUMFERENCE (pi = 3.1416, no extra lib))

```cpp
#include<iostream>
using namespace std;
int main() {
    float r, pi = 3.1416f, area, circ;
    cout << "Enter radius: ";
    cin >> r;
    area = pi * r * r;
    circ = 2 * pi * r;
    cout << "Area          = " << area << " Square Meter" << endl;
    cout << "Circumference = " << circ << " Meter" << endl;
    return 0;
}
```

৬. ধনাত্মক বা ঋণাত্মক সংখ্যা (6. POSITIVE OR NEGATIVE NUMBER)

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    if (n > 0)
        cout << n << " is Positive." << endl;
    else if (n < 0)
        cout << n << " is Negative." << endl;
    else
        cout << "The number is Zero." << endl;
    return 0;
}
```

৭. জোড় বা বিজোড় সংখ্যা (7. ODD OR EVEN)

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    if (n % 2 == 0)
        cout << n << " is Even." << endl;
    else
        cout << n << " is Odd." << endl;
    return 0;
}
```

৮. লিপ ইয়ার (অধিবর্ষ) (8. LEAP YEAR)

```
#include<iostream>
using namespace std;
int main() {
    int year;
    cout << "Enter a year: ";
    cin >> year;
    if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
        cout << year << " is a Leap Year." << endl;
    else
        cout << year << " is not a Leap Year." << endl;
    return 0;
}
```

৯a. সেলসিয়াস থেকে ফারেনহাইট (9a. CELSIUS TO FAHRENHEIT)

```
#include<iostream>
using namespace std;
int main() {
    float c, f;
    cout << "Enter temperature in Celsius: ";
    cin >> c;
    f = (c * 9 / 5) + 32;
    cout << c << " C = " << f << " F" << endl;
    return 0;
}
```

৯b. ফারেনহাইট থেকে সেলসিয়াস (9b. FAHRENHEIT TO CELSIUS)

```
#include<iostream>
using namespace std;
int main() {
    float f, c;
    cout << "Enter temperature in Fahrenheit: ";
    cin >> f;
    c = (f - 32) * 5 / 9;
    cout << f << " F = " << c << " C" << endl;
    return 0;
}
```

১০a. বড় বা ছোট হাতের অক্ষর (Range) (10a. UPPERCASE OR LOWERCASE - METHOD 1 (if-else with range))

```
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a letter: ";
    cin >> ch;
    if (ch >= 'a' && ch <= 'z')
        cout << ch << " is Lowercase." << endl;
    else if (ch >= 'A' && ch <= 'Z')
        cout << ch << " is Uppercase." << endl;
    else
        cout << ch << " is not a letter." << endl;
    return 0;
}
```

১০b. বড় বা ছোট হাতের অক্ষর (ASCII) (10b. UPPERCASE OR LOWERCASE - METHOD 2 (ASCII values))

```
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a letter: ";
    cin >> ch;
    if (ch >= 65 && ch <= 90)
        cout << ch << " is Uppercase. ASCII = " << (int)ch << endl;
    else if (ch >= 97 && ch <= 122)
        cout << ch << " is Lowercase. ASCII = " << (int)ch << endl;
    else
        cout << ch << " is not a letter." << endl;
    return 0;
}
```

১১a. স্বরবর্ণ বা ব্যঞ্জনবর্ণ - if-else (11a. VOWEL OR CONSONANT - if-else METHOD)

```
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a character: ";
    cin >> ch;
    if (ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u'||
        ch=='A'||ch=='E'||ch=='I'||ch=='O'||ch=='U')
        cout << ch << " is a Vowel." << endl;
    else
        cout << ch << " is a Consonant." << endl;
    return 0;
}
```

১১b. স্বরবর্ণ বা ব্যঞ্জনবর্ণ - if-else (11b. VOWEL OR CONSONANT - SWITCH CASE METHOD)

```
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a character: ";
    cin >> ch;
    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            cout << ch << " is a Vowel." << endl;
            break;
        default:
            cout << ch << " is a Consonant." << endl;
    }
    return 0;
}
```

১২a. "Bangladesh" ১০ বার প্রিন্ট - for লুপ (12a. PRINT "Bangladesh" 10 TIMES - FOR LOOP)

```
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++)
        cout << "Bangladesh" << endl;
    return 0;
}
```

১২b. "Bangladesh" ১০ বার প্রিন্ট - while লুপ (12b. PRINT "Bangladesh" 10 TIMES - WHILE LOOP)

```
#include<iostream>
using namespace std;
int main() {
    int i = 1;
    while (i <= 10) {
        cout << "Bangladesh" << endl;
        i++;
    }
    return 0;
}
```

১২c. "Bangladesh" ১০ বার প্রিন্ট - while লুপ (12c. PRINT "Bangladesh" 10 TIMES - DO-WHILE LOOP)

```
#include<iostream>
using namespace std;
int main() {
    int i = 1;
    do {
        cout << "Bangladesh" << endl;
        i++;
    } while (i <= 10);
    return 0;
}
```

১২d. N বার টেক্সট প্রিন্ট - for লুপ (12d. PRINT USER TEXT N TIMES - FOR LOOP)

```
#include<iostream>
#include<string>
using namespace std;
int main() {
    string text;
    int n;
    cout << "Enter text: ";
    cin.ignore();
    getline(cin, text);
    cout << "How many times? ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        cout << text << endl;
    return 0;
}
```

১২e. N বার টেক্সট প্রিন্ট - while লুপ (12e. PRINT USER TEXT N TIMES - WHILE LOOP)

```
#include<iostream>
#include<string>
using namespace std;
int main() {
    string text;
    int n, i = 1;
    cout << "Enter text: ";
    cin.ignore();
    getline(cin, text);
    cout << "How many times? ";
    cin >> n;
    while (i <= n) {
        cout << text << endl;
        i++;
    }
    return 0;
}
```

১২f. N বার টেক্সট প্রিন্ট - do-while লুপ (12f. PRINT USER TEXT N TIMES - DO-WHILE LOOP)

```
#include<iostream>
#include<string>
using namespace std;
int main() {
    string text;
    int n, i = 1;
    cout << "Enter text: ";
    cin.ignore();
    getline(cin, text);
    cout << "How many times? ";
    cin >> n;
    do {
        cout << text << endl;
        i++;
    } while (i <= n);
    return 0;
}
```

১৩a. ১ থেকে ১০ পর্যন্ত প্রিন্ট (13a. PRINT 1 TO 10)

```
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++)
        cout << i << " ";
    cout << endl;
    return 0;
}
```

১৩b. ১ থেকে N পর্যন্ত প্রিন্ট (13b. PRINT 1 TO N (USER INPUT))

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        cout << i << " ";
    cout << endl;
    return 0;
}
```

১৪a. ১ থেকে ১০ এর যোগফল (14a. SUM OF 1 TO 10)

```
#include<iostream>
using namespace std;
int main() {
    int sum = 0;
    for (int i = 1; i <= 10; i++)
        sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

১৪b. ১ থেকে N এর যোগফল (14b. SUM OF 1 TO N (USER INPUT))

```
#include<iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

১৫. মৌলিক সংখ্যা (Prime Number) চেক (15. PRIME NUMBER CHECK)

```
#include<iostream>
using namespace std;
int main() {
    int n, isPrime = 1;
    cout << "Enter a number: ";
    cin >> n;
    if (n < 2) isPrime = 0;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) { isPrime = 0; break; }
    }
    if (isPrime)
        cout << n << " is a Prime number." << endl;
    else
        cout << n << " is not a Prime number." << endl;
    return 0;
}
```

১৬. পারফেক্ট সংখ্যা (Perfect Number) চেক (16. PERFECT NUMBER CHECK)

```
#include<iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 1; i < n; i++)
        if (n % i == 0) sum += i;
    if (sum == n)
        cout << n << " is a Perfect number." << endl;
    else
        cout << n << " is not a Perfect number." << endl;
    return 0;
}
```

১৭. অঙ্কগুলোর যোগফল (17. SUM OF DIGITS)

```
#include<iostream>
using namespace std;
int main() {
    int n, sum = 0, temp;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) {
        sum += temp % 10;
        temp /= 10;
    }
    cout << "Sum of digits of " << n << " = " << sum << endl;
    return 0;
}
```

১৮. সংখ্যাটি উল্টানো (Reverse) (18. REVERSE A NUMBER)

```
#include<iostream>
using namespace std;
int main() {
    int n, rev = 0;
    cout << "Enter a number: ";
    cin >> n;
    while (n != 0) {
        rev = rev * 10 + n % 10;
        n /= 10;
    }
    cout << "Reversed number = " << rev << endl;
    return 0;
}
```

১৯. প্যালিনড্রোম সংখ্যা (19. PALINDROME NUMBER)

```
#include<iostream>
using namespace std;
int main() {
    int n, rev = 0, temp;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) {
        rev = rev * 10 + temp % 10;
        temp /= 10;
    }
    if (rev == n)
        cout << n << " is a Palindrome." << endl;
    else
        cout << n << " is not a Palindrome." << endl;
    return 0;
}
```

২০. আর্মস্ট্রং সংখ্যা (20. ARMSTRONG NUMBER)

```
#include<iostream>
using namespace std;
int main() {
    int n, temp, digits = 0, sum = 0, t;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) { digits++; temp /= 10; }
    temp = n;
    while (temp != 0) {
        t = temp % 10;
        int p = 1;
        for (int i = 0; i < digits; i++) p *= t;
        sum += p;
        temp /= 10;
    }
    if (sum == n)
        cout << n << " is an Armstrong number." << endl;
    else
        cout << n << " is not an Armstrong number." << endl;
    return 0;
}
```

২১. স্ট্রং সংখ্যা (21. STRONG NUMBER)

```
   (sum of factorials of digits = number)
   Example: 145 = 1! + 4! + 5! = 145
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, temp, sum = 0, digit, fact;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) {
        digit = temp % 10;
        fact = 1;
        for (int i = 1; i <= digit; i++) fact *= i;
        sum += fact;
        temp /= 10;
    }
    if (sum == n)
        cout << n << " is a Strong number." << endl;
    else
        cout << n << " is not a Strong number." << endl;
    return 0;
}
```

২২. দুটি সংখ্যার গসাগু এবং লসাগু (22. GCD AND LCM OF TWO NUMBERS)

```cpp
#include<iostream>
using namespace std;
int main() {
    int a, b, x, y, gcd, lcm;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    x = a; y = b;
    while (y != 0) { int t = y; y = x % y; x = t; }
    gcd = x;
    lcm = (a * b) / gcd;
    cout << "GCD = " << gcd << endl;
    cout << "LCM = " << lcm << endl;
    return 0;
}
```

২৩. ফ্যাক্টোরিয়াল (23. FACTORIAL)

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long fact = 1;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 1; i <= n; i++) fact *= i;
    cout << n << "! = " << fact << endl;
    return 0;
}
```

২৪. ধারা: 1! + 2! + ... + n! (24. SERIES: 1! + 2! + 3! + ... + n!)

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0, fact = 1;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        fact *= i;
        sum += fact;
    }
    cout << "1! + 2! + ... + " << n << "! = " << sum << endl;
    return 0;
}
```

২৫. ফ্যাক্টোরিয়াল (25. SERIES: 2! + 4! + 6! + ... + n! (even factorials))

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0, fact;
    cout << "Enter n (even number): ";
    cin >> n;
    for (int i = 2; i <= n; i += 2) {
        fact = 1;
        for (int j = 1; j <= i; j++) fact *= j;
        sum += fact;
    }
    cout << "2! + 4! + ... + " << n << "! = " << sum << endl;
    return 0;
}
```

২৬. ফ্যাক্টোরিয়াল (26. SERIES: 1! + 3! + 5! + ... + n! (odd factorials))

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0, fact;
    cout << "Enter n (odd number): ";
    cin >> n;
    for (int i = 1; i <= n; i += 2) {
        fact = 1;
        for (int j = 1; j <= i; j++) fact *= j;
        sum += fact;
    }
    cout << "1! + 3! + ... + " << n << "! = " << sum << endl;
    return 0;
}
```

২৭a. ফিবোনাচি ধারা - N পদ পর্যন্ত (27a. FIBONACCI SERIES - PRINT TO N TERMS)

```
#include<iostream>
using namespace std;
int main() {
    int n, a = 0, b = 1, c;
    cout << "Enter number of terms: ";
    cin >> n;
    cout << "Series: " << a << " " << b;
    for (int i = 3; i <= n; i++) {
        c = a + b; cout << " " << c; a = b; b = c;
    }
    cout << endl;
    return 0;
}
```

২৭b. ফিবোনাচি ধারা - প্রিন্ট এবং যোগফল (27b. FIBONACCI SERIES - PRINT SERIES AND SUM)

```
#include<iostream>
using namespace std;
int main() {
    int n, a = 0, b = 1, c;
    long long sum = a + b;
    cout << "Enter number of terms: ";
    cin >> n;
    cout << "Series: " << a << " " << b;
    for (int i = 3; i <= n; i++) {
        c = a + b; cout << " " << c;
        sum += c; a = b; b = c;
    }
    cout << "\nSum = " << sum << endl;
    return 0;
}
```

২৮. ধারা: 1 + 2 + ... + n (28. SERIES: 1 + 2 + 3 + ... + n)

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

২৯. ধারা: 1 + 3 + 5 + ... + n (বিজোড়) (29. SERIES: 1 + 3 + 5 + ... + n (odd numbers))

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n (odd): ";
    cin >> n;
    for (int i = 1; i <= n; i += 2) sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩০. ধারা: 2 + 4 + 6 + ... + n (জোড়) (30. SERIES: 2 + 4 + 6 + ... + n (even numbers))

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n (even): ";
    cin >> n;
    for (int i = 2; i <= n; i += 2) sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩১. ধারা: 1^1 + 2^2 + ... + n^n (31. SERIES: 1^1 + 2^2 + 3^3 + ... + n^n)

```
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        long long p = 1;
        for (int j = 0; j < i; j++) p *= i;
        sum += p;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩২. ধারা: 1^2 + 2^2 + ... + n^2 (32. SERIES: 1^2 + 2^2 + 3^2 + ... + n^2)

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) sum += (long long)i * i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৩. ধারা: 1^2 + 3^2 + ... + n^2 (বিজোড়) (33. SERIES: 1^2 + 3^2 + 5^2 + ... + n^2 (odd squares))

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n (odd): ";
    cin >> n;
    for (int i = 1; i <= n; i += 2) sum += (long long)i * i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৪. ধারা: 2^2 + 4^2 + ... + n^2 (জোড়) (34. SERIES: 2^2 + 4^2 + 6^2 + ... + n^2 (even squares))

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0;
    cout << "Enter n (even): ";
    cin >> n;
    for (int i = 2; i <= n; i += 2) sum += (long long)i * i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৫. ধারা: 1 + 2 + 4 + 8 + ... (35. SERIES: 1 + 2 + 4 + 8 + ... (powers of 2, n terms))

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0, term = 1;
    cout << "Enter number of terms: ";
    cin >> n;
    for (int i = 1; i <= n; i++) { sum += term; term *= 2; }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৬. ধারা: 1 + 3 + 9 + 27 + ... (36. SERIES: 1 + 3 + 9 + 27 + ... (powers of 3, n terms))

```
#include<iostream>
using namespace std;
int main() {
    int n; long long sum = 0, term = 1;
    cout << "Enter number of terms: ";
    cin >> n;
    for (int i = 1; i <= n; i++) { sum += term; term *= 3; }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৭. ধারা: 1/1 + 1/2 + 1/3 + ... + 1/n (37. SERIES: 1/1 + 1/2 + 1/3 + ... + 1/n)

```
#include<iostream>
using namespace std;
int main() {
    int n; float sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) sum += 1.0f / i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৮. ধারা: 1/1^2 + 1/2^2 + 1/3^2 + ... + 1/n^2 (38. SERIES: 1/1^2 + 1/2^2 + 1/3^2 + ... + 1/n^2)

```
#include<iostream>
using namespace std;
int main() {
    int n; float sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) sum += 1.0f / ((float)i * i);
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৯. প্যাটার্ন - স্টার বৃদ্ধিক্রম (39. PATTERN - STARS INCREASING)

```
*
**
***
****
*****
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

৪০. প্যাটার্ন - সংখ্যা বৃদ্ধিক্রম (40. PATTERN - NUMBERS INCREASING)

```
1
12
123
1234
12345
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

৪১. প্যাটার্ন - পুনরাবৃত্ত সারি সংখ্যা (41. PATTERN - REPEATED ROW NUMBER)

```
1
22
333
4444
55555
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

৪২. প্যাটার্ন - ১ ০ পর্যায়ক্রমিক (১ দিয়ে শুরু) (42. PATTERN - 1 0 ALTERNATING START 1)

```
1
10
101
1010
10101
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

৪৩. প্যাটার্ন - ১/০ (বিজোড় সারি=১ জোড় সারি=০) (43. PATTERN - ODD ROW=1 EVEN ROW=0)

```
1
00
111
0000
11111
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

৪৪. প্যাটার্ন - পুনরাবৃত্ত বর্ণমালা (44. PATTERN - REPEATED ALPHABET)

```
A
BB
CCC
DDDD
EEEEE
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

৪৫. প্যাটার্ন - বর্ণমালা বৃদ্ধিক্রম (45. PATTERN - ALPHABETS INCREASING)

```
A
AB
ABC
ABCD
ABCDE
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

৪৬. প্যাটার্ন - স্টার হ্রাসক্রম (46. PATTERN - STARS DECREASING)

```
*****
****
***
**
*
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

৪৭. প্যাটার্ন - সংখ্যা হ্রাসক্রম (47. PATTERN - NUMBERS DECREASING)

```
12345
1234
123
12
1
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

৪৮. প্যাটার্ন - পুনরাবৃত্ত সারি হ্রাসক্রম (48. PATTERN - REPEATED DECREASING ROW NUMBER)

```
55555
4444
333
22
1
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

৪৯. প্যাটার্ন - ১ ০ পর্যায়ক্রমিক হ্রাসক্রম (49. PATTERN - 1 0 ALTERNATING DECREASING)

```
10101
1010
101
10
1
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

৫০. প্যাটার্ন - সারি ভিত্তিক ১/০ হ্রাসক্রম (50. PATTERN - ROW BASED 1/0 DECREASING)

```
11111
0000
111
00
1
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

৫১. প্যাটার্ন - বর্ণমালা হ্রাসক্রম (51. PATTERN - ALPHABETS DECREASING)

```
ABCDE
ABCD
ABC
AB
A
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

৫২. প্যাটার্ন - পুনরাবৃত্ত বর্ণমালা (52. PATTERN - REPEATED ALPHABET DECREASING)

```
EEEEE
DDDD
CCC
BB
A
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

৫৩. প্যাটার্ন - স্টার ডায়মন্ড (53. PATTERN - STARS DIAMOND)

```
*
**
***
****
***
**
*
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

৫৪. প্যাটার্ন - সংখ্যা ডায়মন্ড (54. PATTERN - NUMBERS DIAMOND)

```
1
12
123
1234
123
12
1
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

৫৫. প্যাটার্ন - পুনরাবৃত্ত সারি সংখ্যা (55. PATTERN - REPEATED ROW NUMBER DIAMOND)

```
1
22
333
4444
55555
4444
333
22
1
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

৫৬. প্যাটার্ন - ১/০ ডায়মন্ড (১ দিয়ে শুরু) (56. PATTERN - 1/0 START 1 DIAMOND)

```
1
10
101
1010
10101
1010
101
10
1
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

৫৭. প্যাটার্ন - সারি ভিত্তিক ১/০ ডায়মন্ড (57. PATTERN - ROW BASED 1/0 DIAMOND)

```
1
00
111
0000
11111
0000
111
00
1
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

৫৮. প্যাটার্ন - পুনরাবৃত্ত বর্ণমালা (58. PATTERN - REPEATED ALPHABET DIAMOND)

```
A
BB
CCC
DDDD
EEEEE
DDDD
CCC
BB
A
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

৫৯. প্যাটার্ন - বর্ণমালা ক্রম ডায়মন্ড (59. PATTERN - ALPHABET SEQUENCE DIAMOND)

```
A
AB
ABC
ABCD
ABCDE
ABCD
ABC
AB
A
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}


/* NOTE: Pattern 60 is same as 59 */
```

৬০. প্যাটার্ন - বর্ণমালা ক্রম ডায়মন্ড ২ (60. PATTERN - (Same as 59 - Alphabet Sequence Diamond))

```cpp
// See program 59 above
```

৬১. প্যাটার্ন - ডান-প্রান্তিক স্টার (61. PATTERN - RIGHT-ALIGNED STARS)

```
    *
   **
  ***
 ****
*****
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

৬২. প্যাটার্ন - ডান-প্রান্তিক সংখ্যা (62. PATTERN - RIGHT-ALIGNED NUMBERS)

```
    1
   12
  123
 1234
12345
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

৬৩. প্যাটার্ন - ডান-প্রান্তিক পুনরাবৃত্ত সারি (63. PATTERN - RIGHT-ALIGNED REPEATED ROW NUMBER)

```
    1
   22
  333
 4444
55555
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

৬৪. প্যাটার্ন - ডান-প্রান্তিক ১/০ (১ দিয়ে শুরু) (64. PATTERN - RIGHT-ALIGNED 1/0 START 1)

```
    1
   10
  101
 1010
10101
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

৬৫. প্যাটার্ন - ডান-প্রান্তিক সারি ১/০ (65. PATTERN - RIGHT-ALIGNED ROW BASED 1/0)

```
    1
   00
  111
 0000
11111
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

৬৬. প্যাটার্ন - ডান-প্রান্তিক পুনরাবৃত্ত বর্ণমালা (66. PATTERN - RIGHT-ALIGNED REPEATED ALPHABET)

```
    A
   BB
  CCC
 DDDD
EEEEE
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

৬৭. প্যাটার্ন - ডান-প্রান্তিক বর্ণমালা ক্রম (67. PATTERN - RIGHT-ALIGNED ALPHABET SEQUENCE)

```
    A
   AB
  ABC
 ABCD
ABCDE
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

৬৮. প্যাটার্ন - বাম-প্রান্তিক স্টার হ্রাসক্রম (68. PATTERN - LEFT-INDENTED STARS DECREASING)

```
****
 ***
  **
   *
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

৬৯. প্যাটার্ন - বাম-প্রান্তিক সংখ্যা হ্রাসক্রম (69. PATTERN - LEFT-INDENTED NUMBERS DECREASING)

```
12345
 1234
  123
   12
    1
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

৭০. প্যাটার্ন - বাম-প্রান্তিক পুনরাবৃত্ত সারি হ্রাসক্রম (70. PATTERN - LEFT-INDENTED REPEATED ROW DECREASING)

```
55555
 4444
  333
   22
    1
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

৭১. প্যাটার্ন - বাম-প্রান্তিক ১/০ হ্রাসক্রম (71. PATTERN - LEFT-INDENTED 1/0 DECREASING)

```
10101
 1010
  101
   10
    1
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

৭২. প্যাটার্ন - সারি ভিত্তিক ১/০ হ্রাসক্রম (72. PATTERN - LEFT-INDENTED ROW BASED 1/0 DECREASING)

```
11111
 0000
  111
   00
    1
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

৭৩. প্যাটার্ন - পুনরাবৃত্ত বর্ণমালা হ্রাসক্রম (73. PATTERN - LEFT-INDENTED REPEATED ALPHABET DECREASING)

```
EEEEE
 DDDD
  CCC
   BB
    A
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

৭৪. প্যাটার্ন - বাম-প্রান্তিক বর্ণমালা ক্রম হ্রাসক্রম (74. PATTERN - LEFT-INDENTED ALPHABET SEQUENCE DECREASING)

```
ABCDE
 ABCD
  ABC
   AB
    A
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= row; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

৭৫. প্যাটার্ন - সলিড স্টার স্কয়ার (75. PATTERN - SOLID STAR SQUARE)

```
*****
*****
*****
*****
*****
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

৭৬. প্যাটার্ন - সংখ্যা ক্রম স্কয়ার (76. PATTERN - NUMBER SEQUENCE SQUARE)

```
12345
12345
12345
12345
12345
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << col;
        cout << endl;
    }
    return 0;
}
```

৭৭. প্যাটার্ন - সারি সংখ্যা স্কয়ার (77. PATTERN - ROW NUMBER SQUARE)

```
1111
2222
3333
4444
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << row;
        cout << endl;
    }
    return 0;
}
```

৭৮. প্যাটার্ন - ১০১০ কলাম পর্যায়ক্রমিক স্কয়ার (78. PATTERN - 1010 COLUMN ALTERNATING SQUARE)

```
1010
1010
1010
1010
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << (col % 2 == 1 ? 1 : 0);
        cout << endl;
    }
    return 0;
}
```

৭৯. প্যাটার্ন - ১১১১/০০০০ সারি পর্যায়ক্রমিক স্কয়ার (79. PATTERN - 1111/0000 ROW ALTERNATING SQUARE)

```
1111
0000
1111
0000
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        int val = (row % 2 == 1) ? 1 : 0;
        for (int col = 1; col <= n; col++) cout << val;
        cout << endl;
    }
    return 0;
}
```

৮০. প্যাটার্ন - পুনরাবৃত্ত বর্ণমালা (80. PATTERN - REPEATED ALPHABET SQUARE)

```
AAAA
BBBB
CCCC
DDDD
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << (char)('A' + row - 1);
        cout << endl;
    }
    return 0;
}
```

৯০. প্যাটার্ন - বর্ণমালা ক্রম স্কয়ার (90. PATTERN - ALPHABET SEQUENCE SQUARE)

```
ABCD
ABCD
ABCD
ABCD
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= n; col++) cout << (char)('A' + col - 1);
        cout << endl;
    }
    return 0;
}
```

৯১. প্যাটার্ন - ধারাবাহিক সংখ্যা ত্রিভুজ (91. PATTERN - SEQUENTIAL NUMBERS TRIANGLE)

```
1
2 3
4 5 6
7 8 9 10
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, count = 1;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << count++ << " ";
        cout << endl;
    }
    return 0;
}
```

৯২. প্যাটার্ন - ০ থেকে ধারাবাহিক সংখ্যা (92. PATTERN - SEQUENTIAL FROM 0)

```
0
1 2
3 4 5
6 7 8 9
```

```
#include<iostream>
using namespace std;
int main() {
    int n, count = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << count++ << " ";
        cout << endl;
    }
    return 0;
}
```

৯৩. প্যাটার্ন - গুণের নামতা ত্রিভুজ (93. PATTERN - MULTIPLICATION TABLE TRIANGLE)

```
1
2 4
4 6 9
4 8 12 16
(row*col per cell)
```

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int col = 1; col <= row; col++) cout << row * col << " ";
        cout << endl;
    }
    return 0;
}
```

৯৪. প্যাটার্ন - স্টার পিরামিড (94. PATTERN - PYRAMID OF STARS)

```
   *
  ***
 *****
*******
```

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
        for (int sp = 1; sp <= n - row; sp++) cout << " ";
        for (int col = 1; col <= (2 * row - 1); col++) cout << "*";
        cout << endl;
    }
    return 0;
}
```

৯৫. রিকার্শন ব্যবহার করে ফ্যাক্টোরিয়াল (95. FACTORIAL USING RECURSION)

```cpp
#include<iostream>
using namespace std;
long long fact(int n) {
    if (n == 0 || n == 1) return 1;
    return n * fact(n - 1);
}
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    cout << n << "! = " << fact(n) << endl;
    return 0;
}
```

৯৬. রিকার্শন ব্যবহার করে ফিবোনাচি (96. FIBONACCI USING RECURSION)

```
#include<iostream>
using namespace std;
int fib(int n) {
    if (n <= 1) return n;
    return fib(n - 1) + fib(n - 2);
}
int main() {
    int n;
    cout << "Enter number of terms: ";
    cin >> n;
    cout << "Series: ";
    for (int i = 0; i < n; i++) cout << fib(i) << " ";
    cout << endl;
    return 0;
}
```

৯৭. রিকার্শন ব্যবহার করে পাওয়ার (ঘাত) নির্ণয় (97. CALCULATE POWER USING RECURSION (base ^ exp))

```
#include<iostream>
using namespace std;
long long power(int base, int exp) {
    if (exp == 0) return 1;
    return base * power(base, exp - 1);
}
int main() {
    int base, exp;
    cout << "Enter base and exponent: ";
    cin >> base >> exp;
    cout << base << "^" << exp << " = " << power(base, exp) << endl;
    return 0;
}
```

৯৮a. অ্যারে - সর্বোচ্চ মান নির্ণয় (98a. ARRAY - FIND MAXIMUM)

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "How many numbers? ";
    cin >> n;
    int arr[n];
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) cin >> arr[i];
    int max = arr[0];
    for (int i = 1; i < n; i++)
        if (arr[i] > max) max = arr[i];
    cout << "Maximum = " << max << endl;
    return 0;
}
```

৯৮b. অ্যারে - সর্বনিম্ন মান নির্ণয় (98b. ARRAY - FIND MINIMUM)

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "How many numbers? ";
    cin >> n;
    int arr[n];
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) cin >> arr[i];
    int min = arr[0];
    for (int i = 1; i < n; i++)
        if (arr[i] < min) min = arr[i];
    cout << "Minimum = " << min << endl;
    return 0;
}
```

৯৯. অ্যারে - সর্বোচ্চ ও সর্বনিম্ন মান নির্ণয় (99. ARRAY - FIND BOTH MAX AND MIN)

```
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "How many numbers? ";
    cin >> n;
    int arr[n];
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) cin >> arr[i];
    int max = arr[0], min = arr[0];
    for (int i = 1; i < n; i++) {
        if (arr[i] > max) max = arr[i];
        if (arr[i] < min) min = arr[i];
    }
    cout << "Maximum = " << max << endl;
    cout << "Minimum = " << min << endl;
    return 0;
}
```

১০০. টেম্পোরারি ভেরিয়েবল ছাড়া সংখ্যা সোয়াপ (Swap) (100. SWAP TWO NUMBERS WITHOUT TEMP VARIABLE)

```
#include<iostream>
using namespace std;
int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Before: a = " << a << ", b = " << b << endl;
    a = a + b;
    b = a - b;
    a = a - b;
    cout << "After : a = " << a << ", b = " << b << endl;
    return 0;
}
```

১০১a. কন্টিনিউ (Continue) এর ব্যবহার (101a. CONTINUE EXAMPLE)

```
(Print 1-10, skip multiples of 3)
```

```
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
        if (i % 3 == 0) continue;
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

১০১b. ব্রেক (Break) এর ব্যবহার (101b. BREAK EXAMPLE)

```
(Print 1-10, stop when number reaches 6)
```

```
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 6) break;
        cout << i << " ";
    }
    cout << endl;
    return 0;
}
```